



UNIVERSITI TEKNIKAL MALAYSIA MELAKA
PEPERIKSAAN AKHIR SEMESTER II
FINAL EXAMINATION SEMESTER II
SESI 2025/2026
SESSION 2025/2026

FAKULTI KECERDASAN BUATAN DAN KESELAMATAN SIBER

KOD KURSUS : **BAXZ 1313**
COURSE CODE

KURSUS : **KOMUNIKASI DATA DAN RANGKAIAN**
COURSE : **DATA COMMUNICATION AND NETWORKING**

PENYELARAS : **TS. DR. ZULKIFLE MUSLIM**
COORDINATOR

PROGRAM : **BAXI / BAXZ**
PROGRAMME

MASA : **X: XX AGI**
TIME : **X:XX AM**

TEMPOH : **2 JAM**
DURATION : **2 HOURS**

TARIKH :
DATE

TEMPAT : **XXX**
VENUE : **XXX**

ARAHAN KEPADA CALON
INSTRUCTION TO CANDIDATES

1. Kertas soalan ini mengandungi 4 SOALAN SAHAJA. Sila jawab SEMUA soalan.
The exam paper consists of 4 QUESTIONS ONLY. Please answer ALL questions.
 2. Sila jawab SEMUA soalan di dalam buku jawapan yang disediakan.
Please answer ALL questions in the answer booklet provided.
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3 QUESTIOS (56 MARKS) INSTRUCTION: ANSWER ALL QUESTIONS

QUESTION 1 (17 MARKS)

- a) The TCP/IP Model is used as a practical framework for data communication. List 5 (FIVE) item related to Protocol Data Unit (PDU).

(5 marks)

Answer: (1M per item)

1. Data
2. Segment
3. Packet
4. Frame
5. Bits

- b) Consider a noisy channel with a bandwidth of 1 MHz (Hz) and a Signal-to-Noise Ratio (SNR) of 63.

- i) Calculate the Shannon capacity for this channel.

(3 Marks)

- ii) Using the Nyquist formula, find the appropriate number of signal levels () required to achieve this capacity.

(3 Marks)

Answer:

$$a) \quad C = B \log (1 + \text{SNR})$$

$$C = (1 \times 10^6) \times \log (1 + 63)$$

$$C = 6 \text{ Mbps}$$

$$b) \quad \text{Bit Rate} = 2 \times B \times \log L$$

$$6 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log L$$

$$L = 8$$

- c) A periodic signal is decomposed into five sine waves with frequencies of 100 Hz, 300 Hz, 500 Hz, 700 Hz, and 900 Hz. Each component has a maximum amplitude of 10 V. (C3, S)

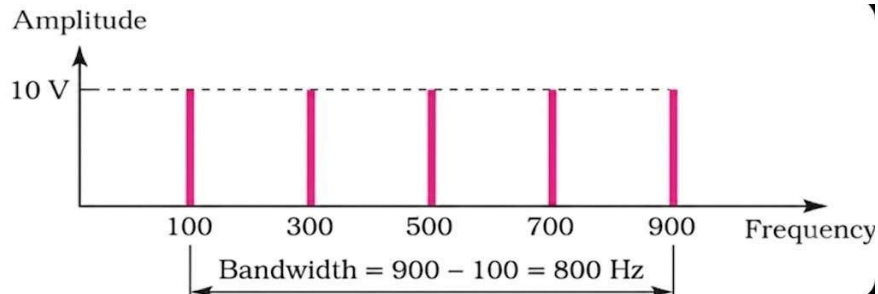
(6 Marks)

- i) Compute the bandwidth of the signal.
- ii) Sketch the frequency spectrum of the signal, clearly labelling the frequency and amplitude of each component.
- iii) Can this signal faithfully pass through a transmission medium that only passes frequencies from 3000 Hz to 4000 Hz? Explain your answer.

Answer:

(i) $\text{Bandwidth} = f_{\text{max}} - f_{\text{min}}$
 $= 900 \text{ Hz} - 100 \text{ Hz}$ (1 mark)
 $= 800 \text{ Hz}$ (1 mark)

(ii)



Answer:

Correct label of 10v (0.5 mark)
 Correct label of Amplitude and Frequency (0.5 mark) Correct label of Bandwidth (0.5 mark)
 Correct label of each frequency value (0.5 mark)

(iii)

Answer:

No (1 mark).

Although the medium has a 1000 Hz bandwidth, the frequency ranges do not overlap, meaning the signal will be totally lost (1 mark)

QUESTION 2 (24 MARKS)

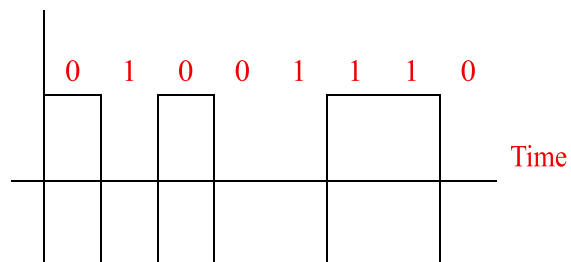
a) A signal is sending a bit stream value of 01001110. Illustrate the produced signal by using:

- i. NRZ-L ii. Differential Manchester Encoding

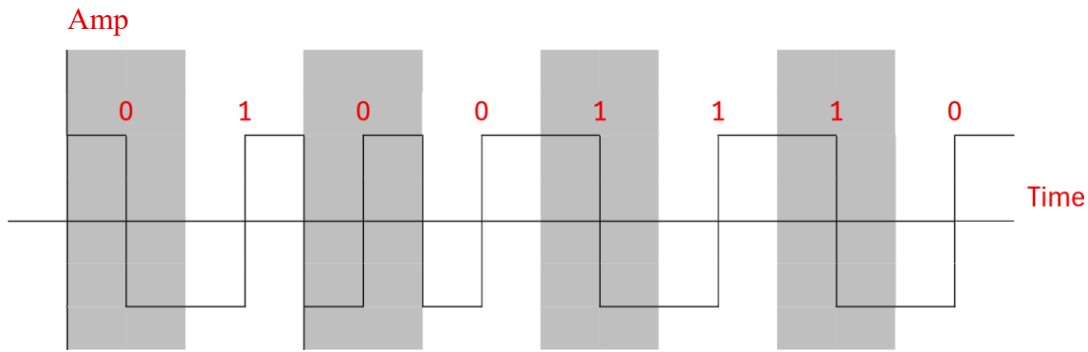
(6 marks)

Answer:

(Correct Shape 2M, Correct Label 1M)
 NRZ-L (3 Marks)



Differential Manchester Encoding (3 Marks)



- b) Describe Define what a Direct-Current (DC) component is in a digital signal and state why it causes problems in transmission.

(3 Marks)

Answer:

When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies (around zero) called DC components (2 marks)

These present problems for a system that cannot pass low frequencies

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FORMATIVE ASSESSMENT

(BAXZ1313)

FORMATIVE ASSESSMENT

- c) An analog signal has a bandwidth of 10,000 Hz (from 1000 Hz to 11,000 Hz) and it is passed through a band-pass to low-pass conversion before sampling. Calculate the required sampling rate based on the Nyquist theorem. Show your calculation.

(5 marks)

Answer:

1. If we change a band-pass signal to a low-pass signal before sampling, the sampling rate is twice the bandwidth. (1 mark)
2. Bandwidth = 10,000 Hz (2 marks)
3. Sampling rate = 20,000samples/s (2 marks)

- d) As a network administrator, you are required to plan a network using Variable Length Subnet Masking (VLSM) technique based on the following department requirements:

- Engineering (DE) – 60 users
- Marketing (DM) – 28 users
- Finance (DF) – 14 users

Using the private addressing scheme of 192.168.10.0/24, identify the following information. Show the calculation work.

- Network Address for each department
- Number of wasted addresses for DE

- The first valid address for DM • The last valid address for DF
- The broadcast address for DM

(10 marks)

Answer:

VLSM Design using 192.168.10.0/24

Step 1: Sort departments by number of users (largest first):

- DE – 60 users → Need /26 (62 hosts)
- DM – 28 users → Need /27 (30 hosts)
- DF – 14 users → Need /28 (14 hosts)

Step 2: VLSM Allocation Table:

Dept	Users	Subnet	Network Address	First Valid	Last Valid	Broadcast
DE	60	/26	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
DM	28	/27	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
DF	14	/28	192.168.10.96	192.168.10.97	192.168.10.110	192.168.10.111

Answers with marks:

- Network Address for each department: DE – 192.168.10.0/26 | DM – 192.168.10.64/27 | DF – 192.168.10.96/28 (3M with correct working)
- Wasted addresses for DE: Total hosts = 62; Used = 60; Wasted = 62 – 60 = 2 addresses (1M)
- First valid address for DM: 192.168.10.65/27 (2M)
- Last valid address for DF: 192.168.10.110/28 (2M)
- Broadcast address for DM: 192.168.10.95/27 (2M)

QUESTION 3 (15 MARKS)

- a) State TWO (2) primary advantages of IPv6's security features or header structure compared to IPv4.

(2 marks)

Answer: 1M per answer

- Native IPsec Integration
- Simplified Fixed Header Format
- No Router-Level Fragmentation

- b) An IPv6 address is represented in a shortened format as FE80::2AA:FF:FE00:A1. Expand this address into its full unabbreviated 32-digit hexadecimal version.

(3 marks)

Answer: 1M for answer 2M for the conversion work

FE80:0000:0000:0000:02AA:00FF:FE00:00A1

(Explanation: Each block must have 4 hexadecimal digits, and the double colon "::" is replaced by zero blocks until the full 8 blocks/128 bits are restored.)

- c) Briefly explain the primary role of Guard Bands in a multiplexing system.

(2 marks)

Answer

Guard bands act as narrow, unused buffer frequency ranges deployed between adjacent channels in FDM to prevent crosstalk / signal interference from leaking into neighboring channels

- d) Discuss ONE (1) primary advantage of the Selective-Repeat ARQ protocol compared to the Go-Back-N ARQ protocol.

(2 marks)

Answer:

2M for the explanation

High Bandwidth Efficiency: Selective-Repeat retransmits exclusively the specific corrupted or missing frame. It caches and preserves out-of-order packets that arrived successfully after the error point, eliminating the wasteful bulk retransmissions required by Go-Back-N.

- e) Error Detection Calculation: A node wants to transmit the bit pattern 1101011. Using a generator polynomial representing the divisor 1001, calculate the resulting Cyclic Redundancy Check (CRC) remainder.

(6 marks)

•Data: 1101011 | Divisor: 1001 (add three '0' because divisor was 4-bit: 1101011000)

•Step: Perform XOR division continuously using binary division.

•Remainder (CRC): 111

Calculation of binary division:

1 1 0 0 1 1 1

$$\begin{array}{r}
 10011101011 \overline{)000} \\
 \underline{1001} \\
 0100 \\
 \underline{1001} \\
 0001 \\
 \underline{0000} \\
 0000 \\
 \underline{011} \\
 0000 \\
 \underline{111} \\
 1001 \\
 \underline{111} \\
 1001 \\
 \underline{111} \\
 1001 \\
 \underline{0111} \\
 0111
 \end{array}$$

[4 marks]

0

0

0

(Remainder)

* Calculation 4 marks

Remainder: 111 [1 marks]

There is an error in the data unit: [1 mark]